TUTORIAL – 15 AUTOMATED LOGIN TESTING

Code:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.common.exceptions import (
    NoSuchElementException,
    TimeoutException,
    ElementNotInteractableException,
    StaleElementReferenceException
)

URL = "https://www.saucedemo.com/"
VALID_USERNAME = "standard_user"
VALID_PASSWORD = "secret_sauce"



def start_browser():
    driver = webdriver.Chrome()
    driver.maximize_window()
    return driver



def login(driver, username, password):
    driver.get(URL)
    wait = WebDriverWait(driver, 10)

    username_box = wait.until(
        EC.visibility_of_element_located(
            (By.ID, "user-name")
        )
    )
    password_box = wait.until(
        EC.visibility_of_element_located(
            (By.ID, "password")
        )
    )
    login_button = wait.until(
        EC.element_to_be_clickable(
            (By.ID, "login-button")
        )
    )

    username_box.clear()
    username_box.send_keys(username)
    password_box.clear()
    password_box.send_keys(password)
    login_button.click()



# TC-01 VALID LOGIN
def test_valid_login():
    driver = start_browser()
    try:
        login(
            driver,
```

```
            VALID_USERNAME,
            VALID_PASSWORD
        )
        WebDriverWait(driver, 10).until(
            EC.url_contains("inventory")
        )
        assert "inventory" in driver.current_url
        print("TC-01 Valid Login: PASS")
    except AssertionError:
        print("TC-01 Valid Login: FAIL")
    except NoSuchElementException:
        print("NoSuchElementException handled")
    except TimeoutException:
        print("TimeoutException handled")
    except ElementNotInteractableException:
        print(
            "ElementNotInteractableException handled"
        )
    except StaleElementReferenceException:
        print(
            "StaleElementReferenceException handled"
        )
    finally:
        driver.quit()


# TC-02 INVALID PASSWORD
def test_invalid_password():
    driver = start_browser()
    try:
        login(
            driver,
            VALID_USERNAME,
            "wrong_password"
        )
        error_message = WebDriverWait(
            driver, 10
        ).until(
            EC.visibility_of_element_located(
                (
                    By.CSS_SELECTOR,
                    "h3[data-test='error']"
                )
            )
        )
        assert (
            "Username and password do not match"
            in error_message.text
        )
        print("TC-02 Invalid Password: PASS")
    except AssertionError:
        print("TC-02 Invalid Password: FAIL")
    except TimeoutException:
        print("Error message not displayed")
    finally:
        driver.quit()

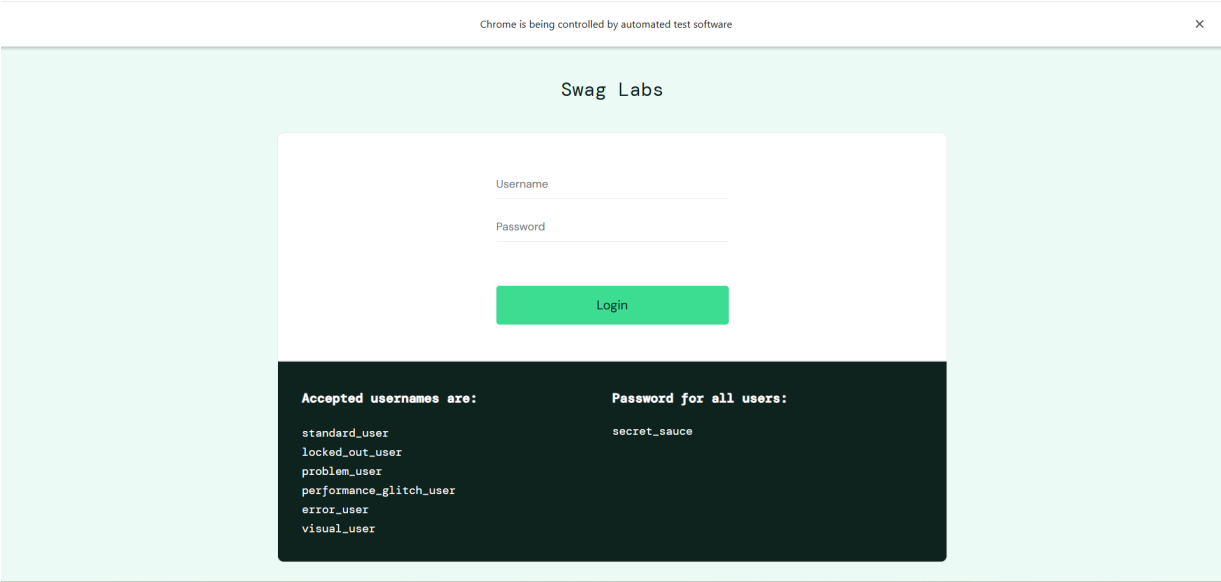

# TC-03 INVALID USERNAME
def test_invalid_username():
    driver = start_browser()
    try:
```

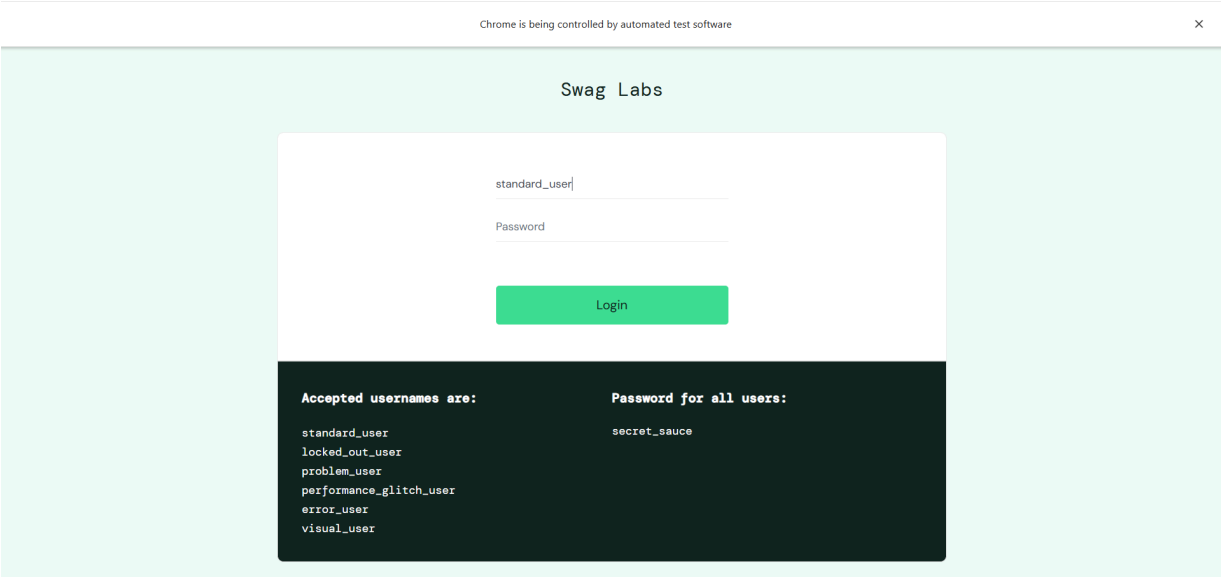
```
        login(
            driver,
            "wrong_user",
            VALID_PASSWORD
        )
        error_message = WebDriverWait(
            driver, 10
        ).until(
            EC.visibility_of_element_located(
                (
                    By.CSS_SELECTOR,
                    "h3[data-test='error']"
                )
            )
        )
        assert (
            "Username and password do not match"
            in error_message.text
        )
        print("TC-03 Invalid Username: PASS")
    except AssertionError:
        print("TC-03 Invalid Username: FAIL")
    except TimeoutException:
        print("Error message not displayed")
    finally:
        driver.quit()


# TC-04 EMPTY CREDENTIALS
def test_empty_credentials():
    driver = start_browser()
    try:
        driver.get(URL)
        login_button = WebDriverWait(
            driver, 10
        ).until(
            EC.element_to_be_clickable(
                (By.ID, "login-button")
            )
        )
        login_button.click()
        error_message = WebDriverWait(
            driver, 10
        ).until(
            EC.visibility_of_element_located(
                (
                    By.CSS_SELECTOR,
                    "h3[data-test='error']"
                )
            )
        )
        assert (
            "Username is required"
            in error_message.text
        )
        print("TC-04 Empty Credentials: PASS")
    except AssertionError:
        print("TC-04 Empty Credentials: FAIL")
    except TimeoutException:
        print("Validation message not displayed")
    finally:
        driver.quit()
```

```
# TC-05 LOGOUT
def test_logout():
    driver = start_browser()
    try:
        login(
            driver,
            VALID_USERNAME,
            VALID_PASSWORD
        )
        WebDriverWait(driver, 10).until(
            EC.url_contains("inventory")
        )
        menu_button = WebDriverWait(
            driver, 10
        ).until(
            EC.element_to_be_clickable(
                (By.ID, "react-burger-menu-btn")
            )
        )
        menu_button.click()
        logout_button = WebDriverWait(
            driver, 10
        ).until(
            EC.element_to_be_clickable(
                (By.ID, "logout_sidebar_link")
            )
        )
        logout_button.click()
        WebDriverWait(driver, 10).until(
            EC.visibility_of_element_located(
                (By.ID, "login-button")
            )
        )
        print("TC-05 Logout: PASS")
    except TimeoutException:
        print("Logout timed out")
    except AssertionError:
        print("TC-05 Logout: FAIL")
    finally:
        driver.quit()
```

Output Screenshots:





Chrome is being controlled by automated test software

Swag Labs

standard_user

Login

Accepted usernames are:

standard_user
locked_out_user
problem_user
performance_glitch_user
error_user
visual_user

Password for all users:

secret_sauce

Chrome is being controlled by automated test software

Swag Labs

Products

Name (A to Z)



Sauce Labs Backpack

carry.allTheThings() with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection.

$29.99 Add to cart



Sauce Labs Bike Light

A red light isn't the desired state in testing but it sure helps when riding your bike at night. Water-resistant with 3 lighting modes, 1 AAA battery included.

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Sauce Labs Bolt T-Shirt

Get your testing superhero on with the Sauce Labs bolt T-shirt. From American Apparel, 100% ringspun combed cotton, heather gray with red bolt.

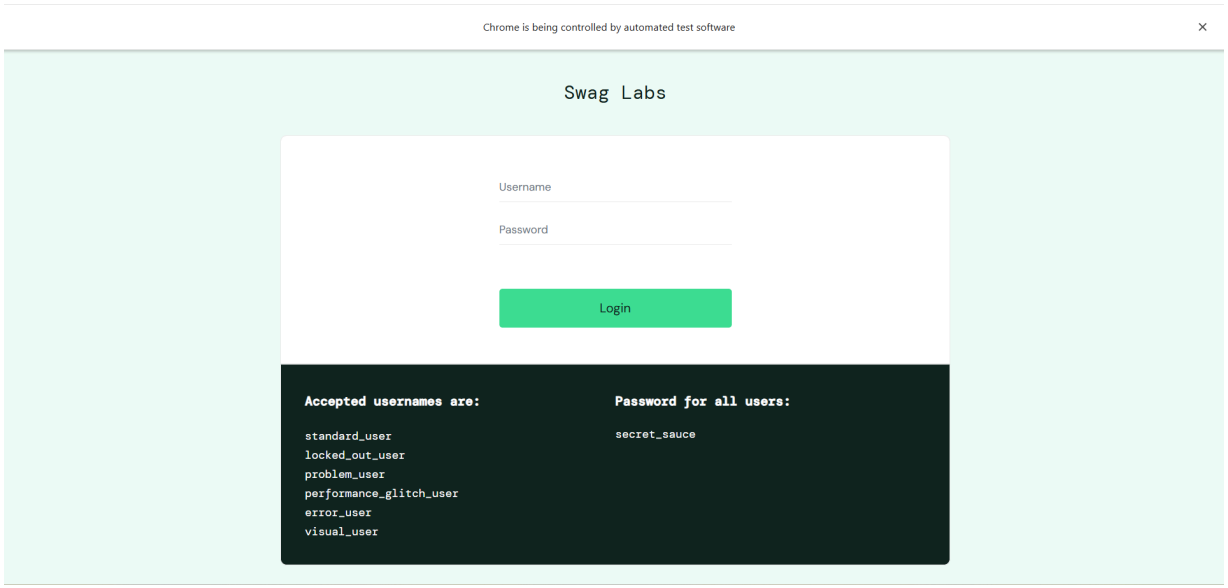
$15.99 Add to cart

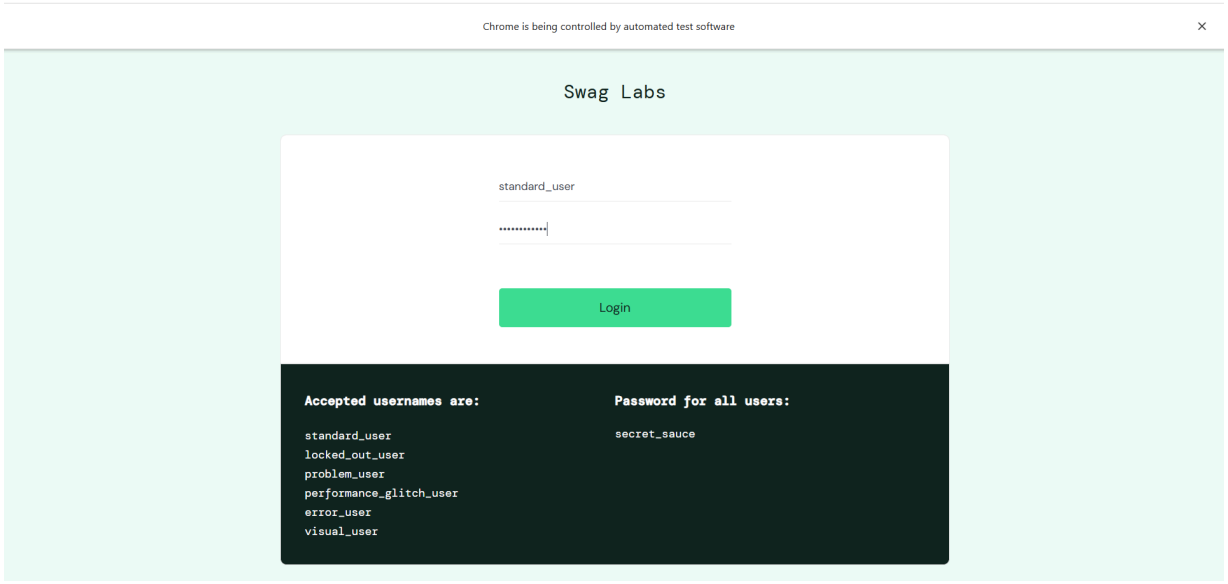


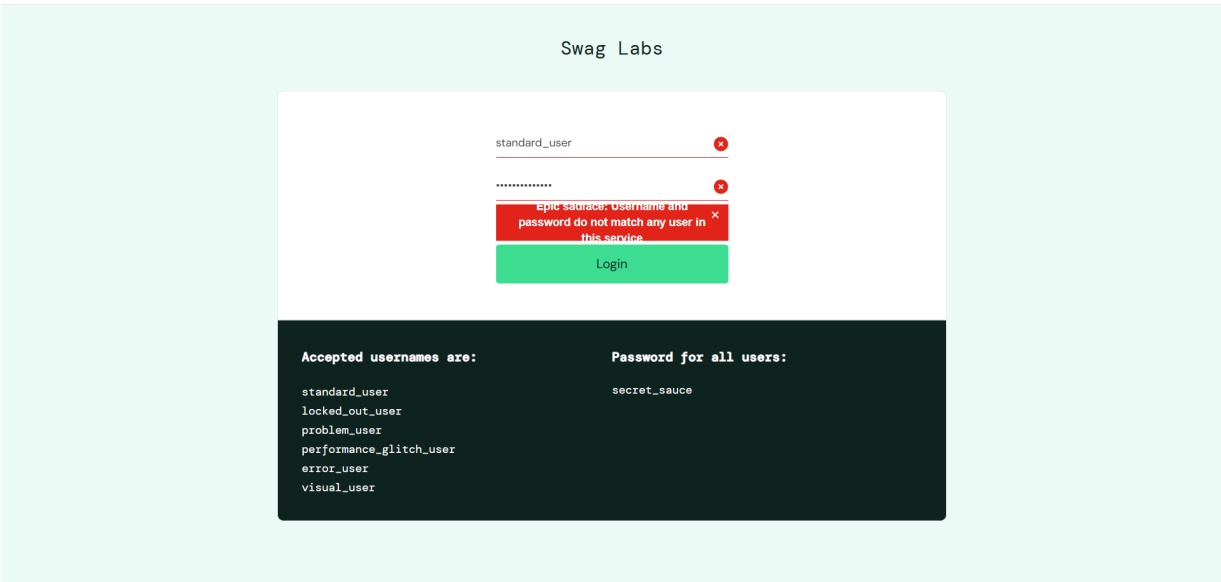
Sauce Labs Fleece Jacket

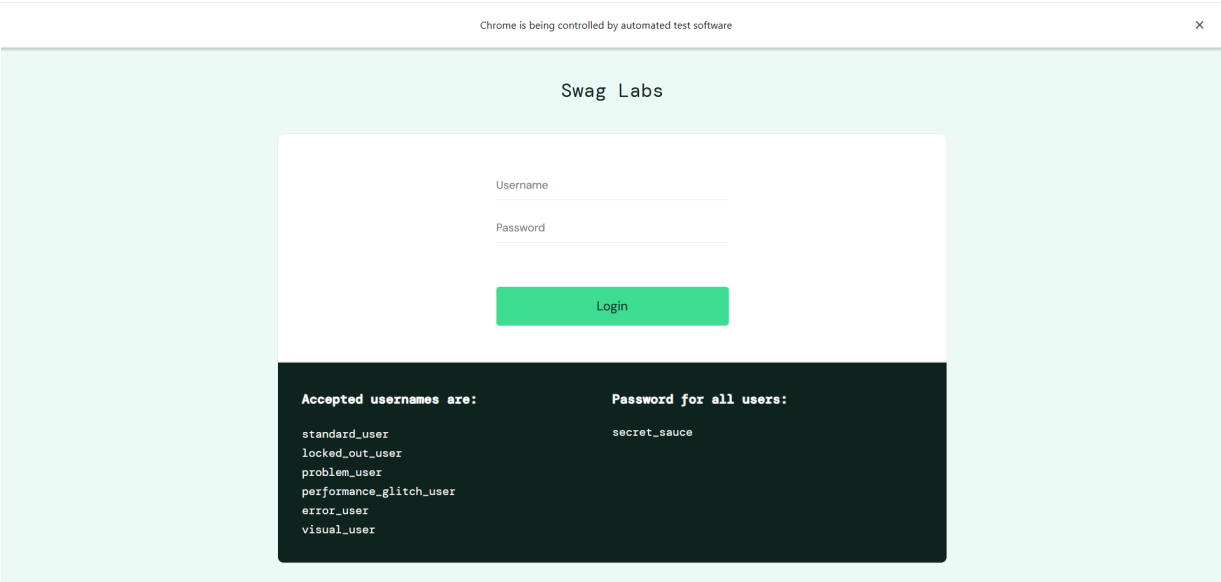
It's not every day that you come across a midweight quarter-zip fleece jacket capable of handling everything from a relaxing day outdoors to a busy day at the office.

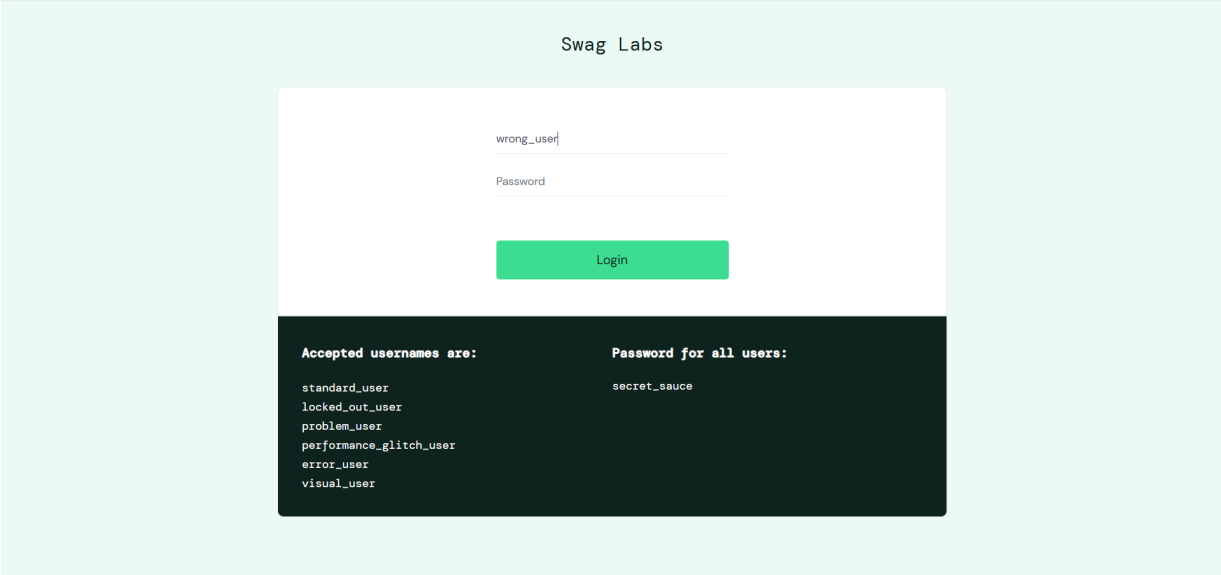
$49.99 Add to cart

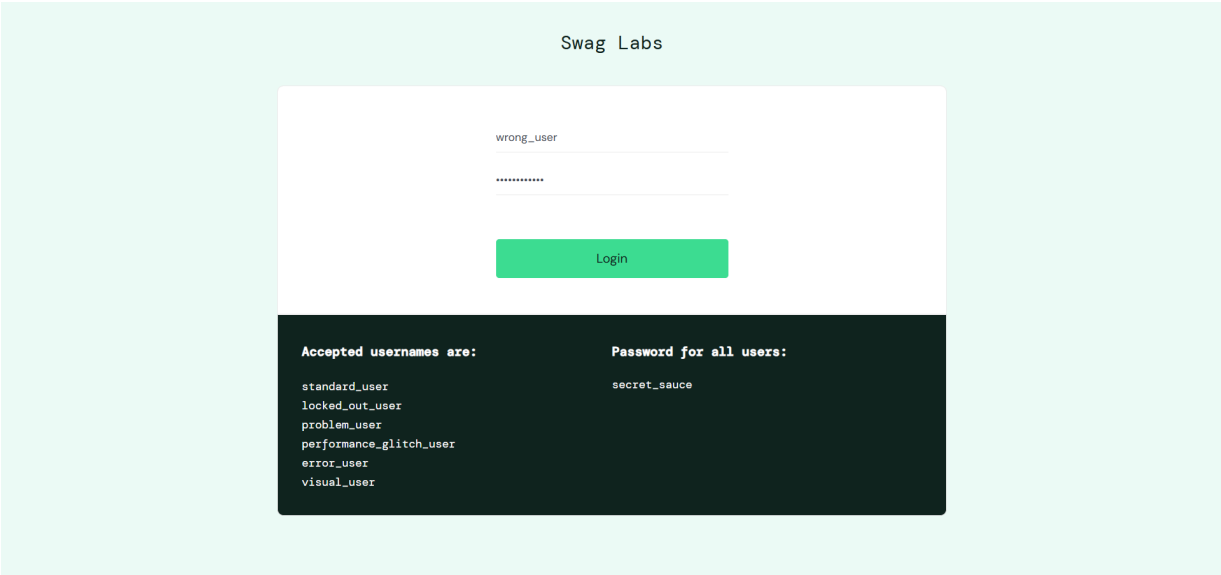

Chrome is being controlled by automated test software
Swag Labs
Username
Password
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Accepted usernames are:
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performance_glitch_user
error_user
visual_user
Password for all users:
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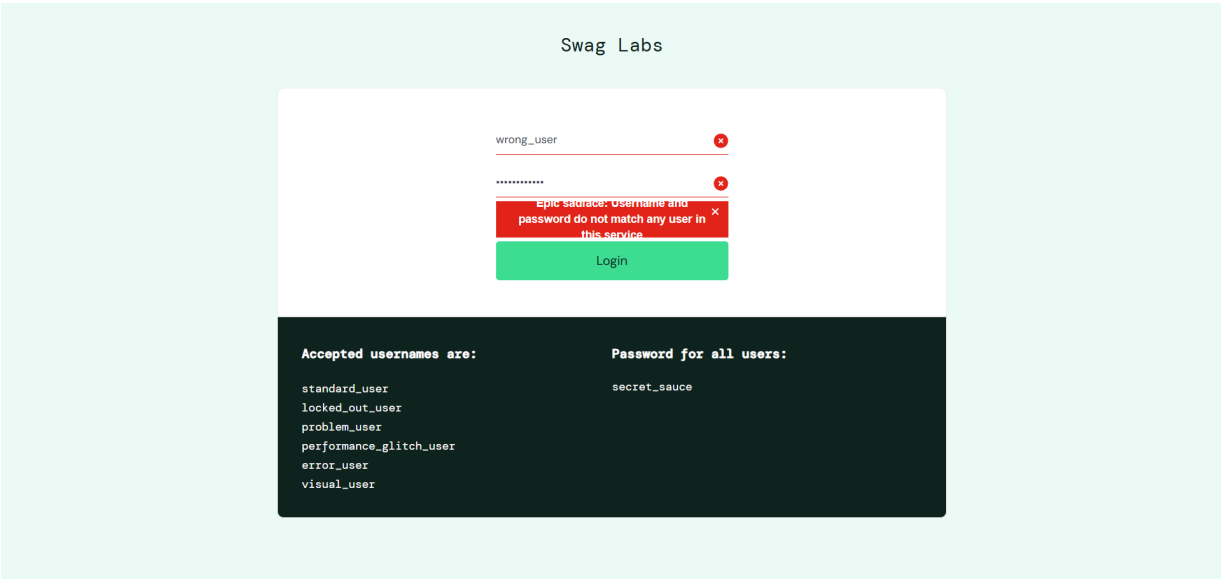

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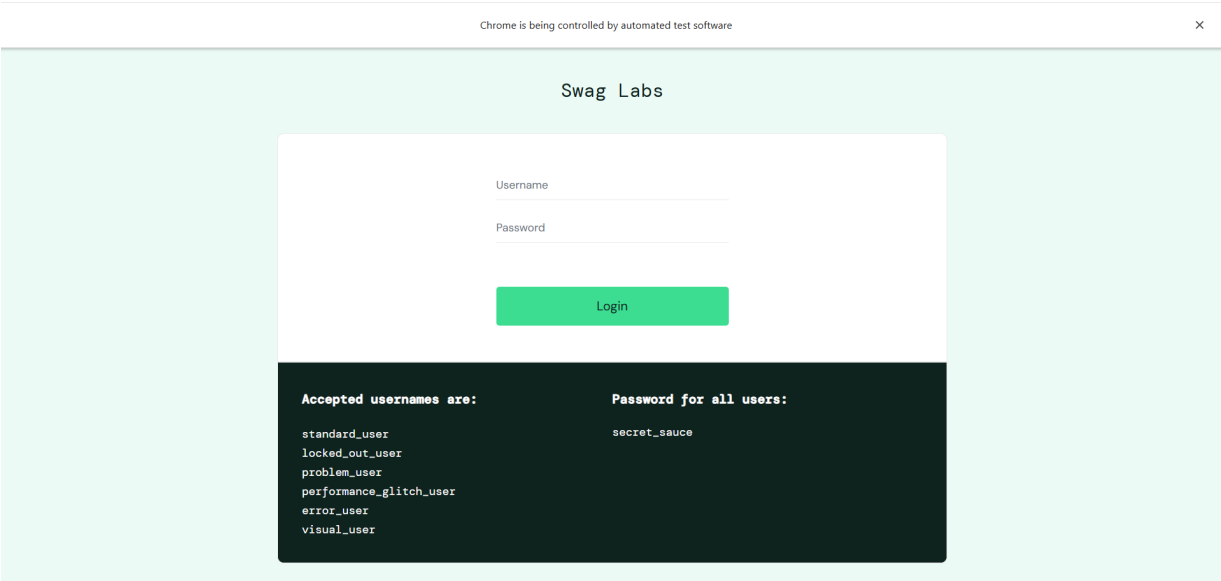

Swag Labs
standard_user
Epic sadface: Username and password do not match any user in this service
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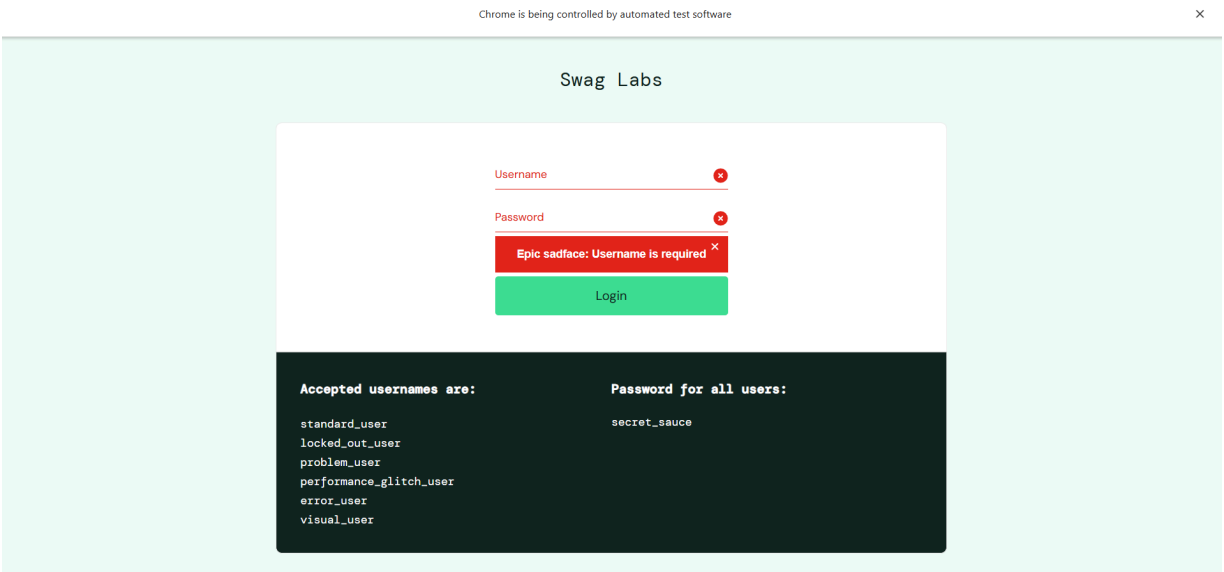

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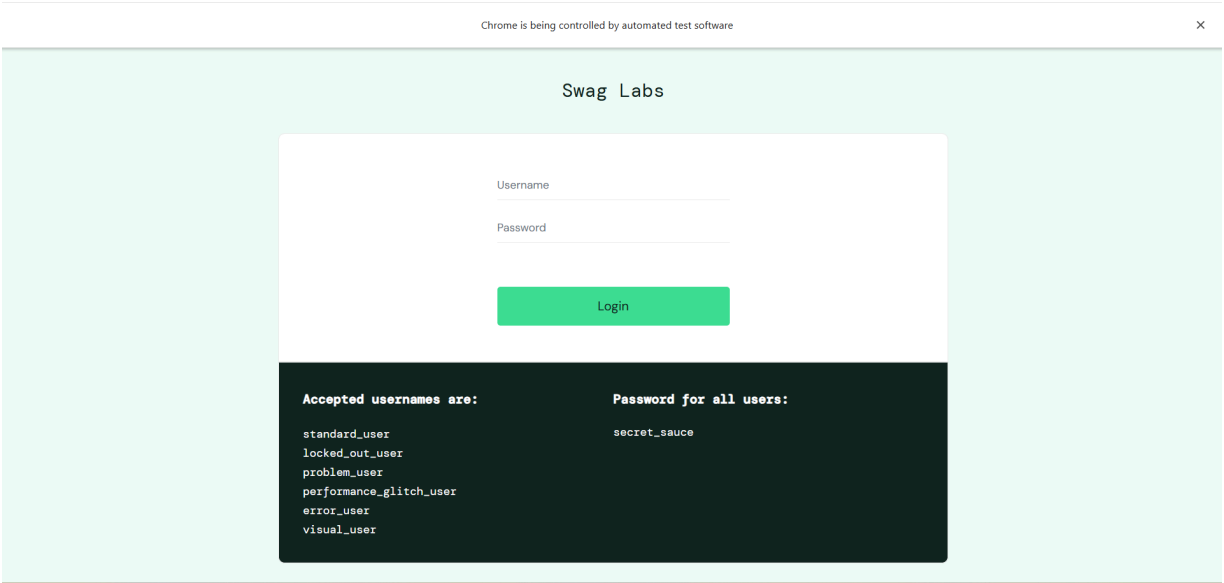

Swag Labs
wrong_user
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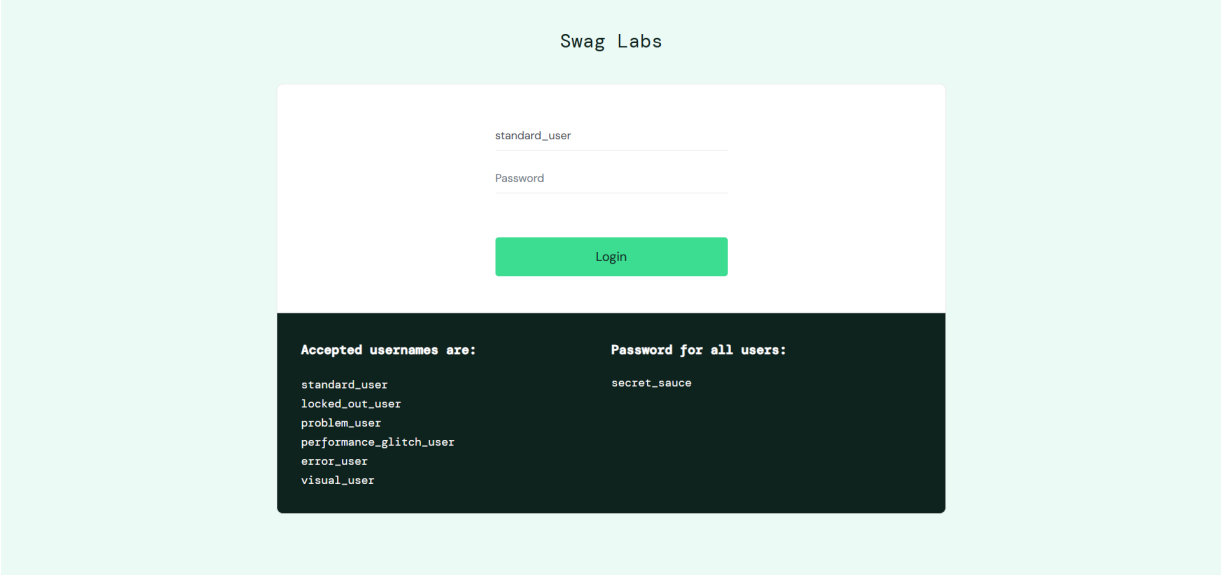

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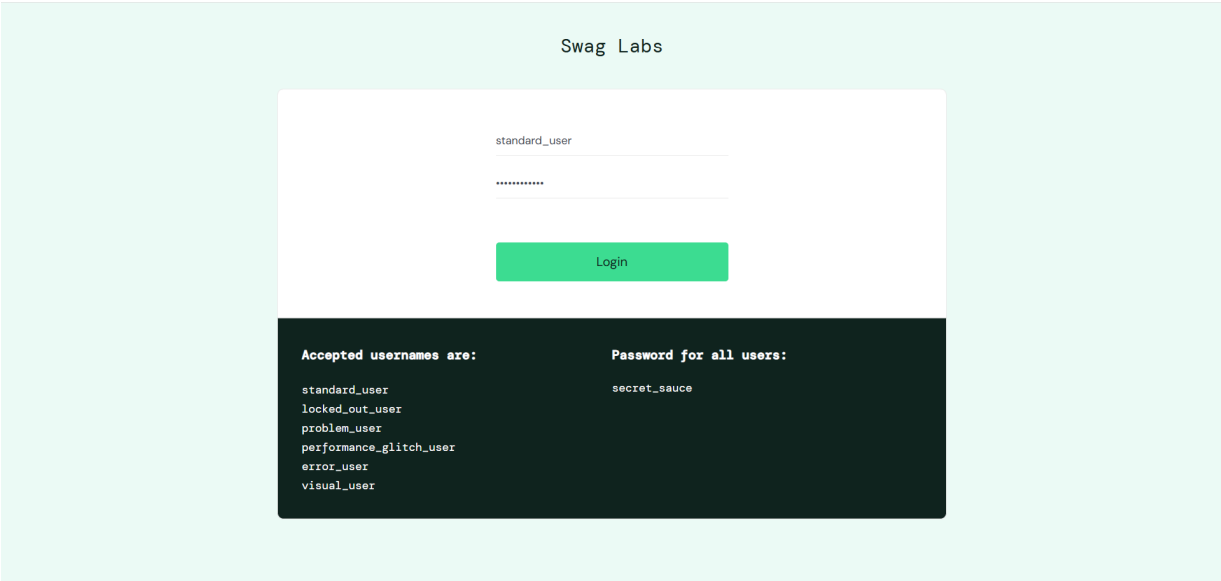

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Products

Name (A to Z)



Sauce Labs Backpack

carry.allTheThings() with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection.

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Sauce Labs Bike Light

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It's not every day that you come across a midweight quarter-zip fleece jacket capable of handling everything from a relaxing day outdoors to a busy day at the office.

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Sauce Labs Onesie

Rib snap infant onesie for the junior



Test.allTheThings() T-Shirt (Red)

Swag Labs

Name (A to Z)

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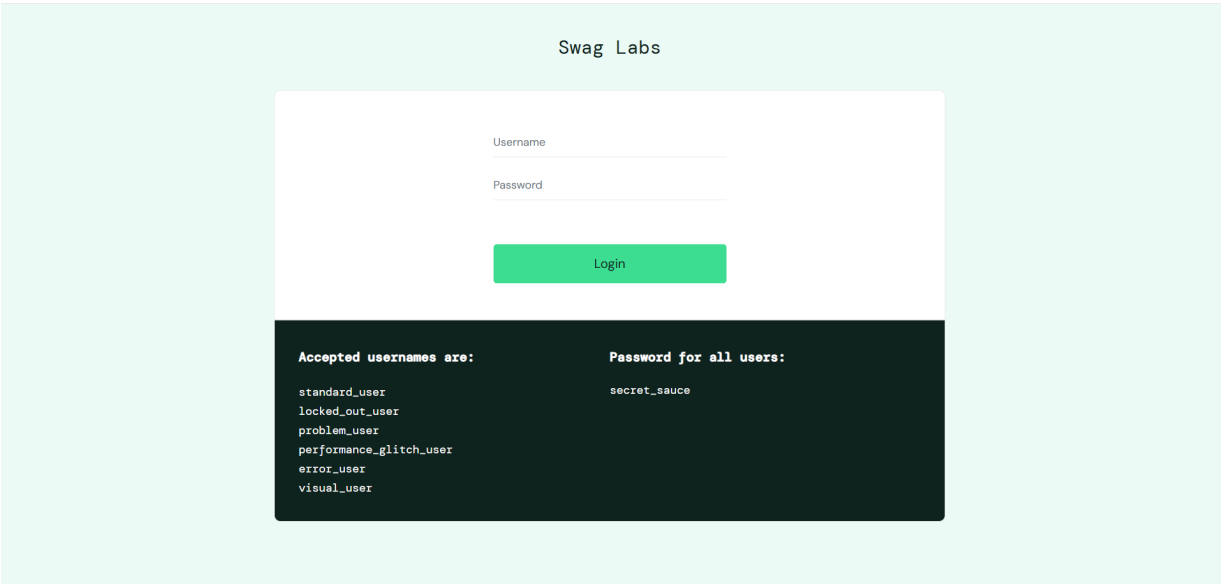
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